JEE(ADVANCED) – 2014 PAPER-1 Code-1

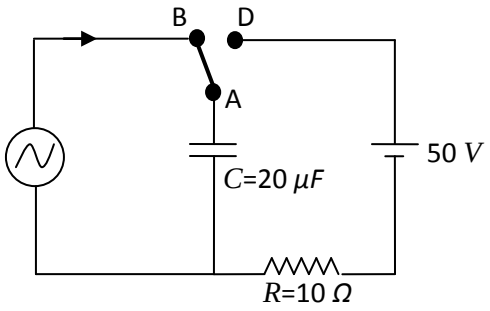
Questions with Answers

PART – 1 PHYSICS

SECTION – 1 (One or More Than One Options Correct Type)

This section contains 10 multiple choice type questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE THAN ONE are correct.

1. At time $t = 0$, terminal A in the circuit shown in the figure is connected to B by a keyand an alternating current $I(t) = I_0 \cos(\omega t)$, with $I_0 = 1A$and ω = 500 *rad* s^{-1} starts flowing in it with the initial direction shown in the figure. At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If C=20μF, R= 10 Ω and the battery is ideal with *emf* of 50V, identify the correct statement (s).



(A) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is $1 \times 10^{-3} C$.

(B) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$is clockwise.

(C) Immediately after A is connected to D, the current in R is 10A.

(D) $Q = 2 \times 10^{-3} C$.

Answer (C) and (D)

2. A light source, which emits two wavelengths $\lambda_1 = 400\,nm$ and $\lambda_2 = 600\,nm$, is used in a Young's double slit experiment. If recorded fringe widths for λ_1 and λ_2 are β_1 and β_2 © number of fringes for them within a distance *y* on one side of the central maximum are m_1 and m_2, respectively, then

(A) $\beta_2 > \beta_1$

(B) $m_1 > m_2$

(C) From the central maximum, 3rd maximum of λ_2 overlaps with 5th minimum of λ_1

(D) The angular separation of fringes for λ_1 is greater than λ_2

Answer: (A), (B) and (C)

3. One end of a taut string of length 3*m* along the *x* axis is fixed at *x*=0.The speed of the waves in the string is $100\,ms^{-1}$. The other end of the string is vibrating in the *y* direction so that stationary waves are set up in the string. The possible waveform(s) of these stationary waves is(are)

(A) $y(t) = A\sin\frac{\pi x}{6}\cos\frac{50\pi t}{3}$

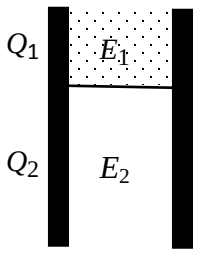
(B) $y(t) = A\sin\frac{\pi x}{3}\cos\frac{100\pi t}{3}$

(C) $y(t) = A\sin\frac{5\pi x}{6}\cos\frac{250\pi t}{3}$

(D) $y(t) = A\sin\frac{5\pi x}{2}\cos 250\pi t$

Answer: (A) , (C) and (D)

4. A parallel plate capacitor has a dielectric slab of dielectric constant *K* between its plates that covers 1/3 of the area of its plates, as shown in the figure. The total capacitance of the capacitor is *C* while that of the portion with dielectric in between is C_1. When the capacitor is charged, the plate area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2. The electric field in the dielectric is E_1 and that in the other portion is E_2. Choose the correct option/options, ignoring edge effects.



(A) $\frac{E_1}{E_2} = 1$

(B) $\frac{E_1}{E_2} = \frac{1}{K}$

(C) $\frac{Q_1}{Q_2} = \frac{3}{K}$

(D) $\frac{C}{C_1} = \frac{2+K}{K}$

Answer: (A) and (D)

5. Let $E_1(r)$, $E_2(r)$ and $E_3(r)$ be the respective electric fields at a distance *r* from a point charge *Q*, an infinitely long wire with constant linear charge density λ, and an infinite plane with uniform surface charge density σ. If $E_1(r_0) = E_2(r_0) = E_3(r_0)$ at a given distance r_0, then

(A) $Q = 4\sigma\pi r_0^2$

(B) $r_0 = \frac{\lambda}{2\pi\sigma}$

(C) $E_1(r_0/2) = 2E_2(r_0/2)$

(D) $E_2(r_0/2) = 4E_3(r_0/2)$

Answer: (C)

6. A student is performing an experiment using a resonance column and a tuning fork of frequency $244 s^{-1}$. He is told that the air in the tube has been replaced by another gas (assume that the column remains filled with the gas). If the minimum height at which resonance occurs is $(0.350 \pm 0.005)\ m$, the gas in the tube is

(**Useful information:** $\sqrt{167RT} = 640\ J^{1/2} mole^{-1/2}$; $\sqrt{140RT} = 590\ J^{1/2} mole^{-1/2}$. The molar masses *M* in grams are given in the options. Take the values of $\sqrt{\frac{10}{M}}$ for each gas as given there.)

(A) Neon $\left(M = 20, \sqrt{\frac{10}{20}} = \frac{7}{10}\right)$

(B) Nitrogen $\left(M = 28, \sqrt{\frac{10}{28}} = \frac{3}{5}\right)$

(C) Oxygen $\left(M = 32, \sqrt{\frac{10}{32}} = \frac{9}{16}\right)$

(D) Argon $\left(M = 36, \sqrt{\frac{10}{36}} = \frac{17}{32}\right)$

Answer: (D)

7. Heater of an electric kettle is made of a wire of length *L* and diameter *d*. It takes 4 minutes to raise the temperature of 0.5*kg* water by 40*K*. This heater is replaced by a new heater having two wires of the same material, each of length *L* and diameter 2*d*. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40 *K*?

(A) 4 if wires are in parallel

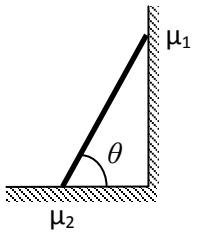
(B) 2 if wires are in series

(C) 1 if wires are in series

(D) 0.5 if wires are in parallel

Answer : (B) and (D)

8. In the figure, a ladder of mass m is shown leaning against a wall. It is in static equilibrium making an angle θ with the horizontal floor. The coefficient of friction between the wall and the ladder is μ_1 and that between the floor and the ladder is μ_2. The normal reaction of the wall on the ladder is N_1 and that of the floor is N_2. If the ladder is about to slip, then



(A) $\mu_1 = 0 \quad \mu_2 \neq 0 \text{and} N_2 \tan\theta = \frac{mg}{2}$

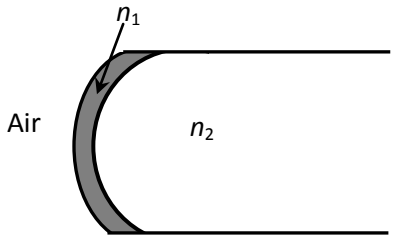
(B) $\mu_1 \neq 0 \quad \mu_2 = 0 \text{and} N_1 \tan\theta = \frac{mg}{2}$

(C) $\mu_1 \neq 0 \quad \mu_2 \neq 0 \text{and} N_2 = \frac{mg}{1+\mu_1\mu_2}$

(D) $\mu_1 = 0 \quad \mu_2 \neq 0 \text{and} N_1 \tan\theta = \frac{mg}{2}$

Answer: (C) and (D)

9. A transparent thin film of uniform thickness and refractive index n_1= 1.4 is coated on the convex spherical surface of radius Rat one end of a long solid glass cylinder of refractive index n_2 = 1.5, as shown in the figure. Rays of light parallel to the axis of the cylinder traversing through the film from air to glass get focused at distance f_1 from the film, while rays of light traversing from glass to air get focused at distance f_2 from the film. Then



(A) $|f_1| = 3R$

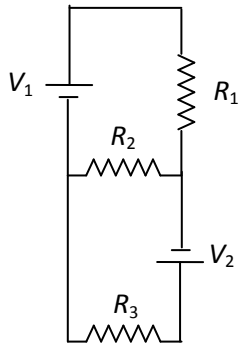
(B) $|f_1| = 2.8R$

(C) $|f_2| = 2R$

(D) $|f_2| = 1.4R$

Answer: (A) and (C)

10. Two ideal batteries of *emf* V_1 and V_2 and three resistances R_1, R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if



(A) $V_1 = V_2 \text{and} R_1 = R_2 = R_3$

(B) $V_1 = V_2 \text{and} R_1 = 2R_2 = R_3$

(C) $V_1 = 2V_2 \text{and } 2R_1 = 2R_2 = R_3$

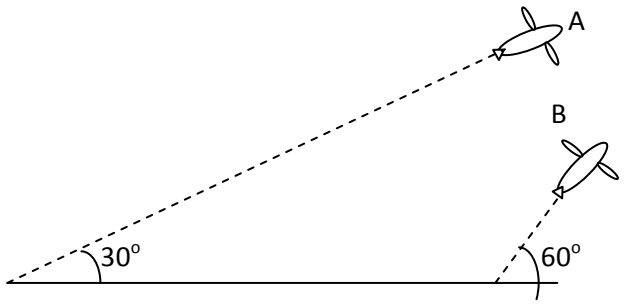
(D) $2V_1 = V_2 \text{and} 2R_1 = R_2 = R_3$

Answer: (A), (B) and (D)

SECTION – 2 (One Integer Value Correct Type)

This section contains 10 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

11. Airplanes A and B are flying with constant velocity in the same vertical plane at angles 30° and 60° with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3}\,ms^{-1}$. At time $t = 0\,s$, an observer in A finds B at a distance of 500 m. This observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$, A just escapes being hit by B, t_0 in seconds is

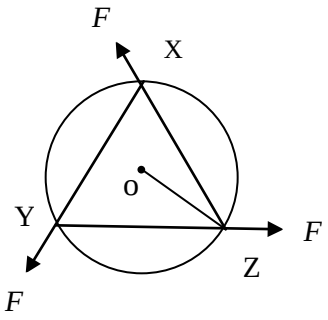


Answer: 5

12. During Searle's experiment, zero of the Vernier scale lies between $3.20 \times 10^{-2}\,m$ and $3.25 \times 10^{-2}\,m$ of the main scale. The 20th division of the Vernier scale exactly coincides with one of the main scale divisions. When an additional load of 2 kg is applied to the wire, the zero of the Vernier scale still lies between $3.20 \times 10^{-2}\,m$ and $3.25 \times 10^{-2}\,m$ of the main scale but now the 45th division of Vernier scale coincides with one of the main scale divisions. The length of the thin metallic wire is 2 m and its cross-sectional area is $8 \times 10^{-7} m^2$. The least count of the Vernier scale is $1.0 \times 10^{-5}\,m$. The maximum percentage error in the Young's modulus of the wire is

Answer: 4

13. A uniform circular disc of mass 1.5 *kg* and radius 0.5 *m* is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude $F = 0.5\ N$ are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in $rads^{-1}$ is



Answer: 2

14. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed *u* between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude *I* in the same direction, the radius of curvature of the path of the point charge is R_1. In contrast, if the currents *I* in the two wires have directions opposite to each other, the radius of curvature of the path is R_2. If $\frac{X_0}{X_1} = 3$, the value of $\frac{R_1}{R_2}$ is

Answer: 3

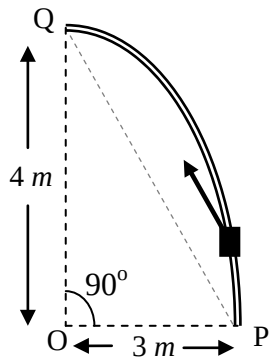
15. To find the distance *d* over which a signal can be seen clearly in foggy conditions, a railways engineer uses dimensional analysis and assumes that the distance depends on the mass density ρ of the fog, intensity (power/area) S of the light from the signal and its frequency *f*. The engineer finds that $d\text{isproportionalto}S^{1/n}$. The value of n is

Answer: 3

16. A galvanometer gives full scale deflection with 0.006 *A* current. By connecting it to a 4990 Ω resistance, it can be converted into a voltmeter of range 0 – 30 *V*. If connected to a $\frac{2n}{249}\Omega$ resistance, it becomes an ammeter of range 0 – 1.5 *A*. The value of$n$ is

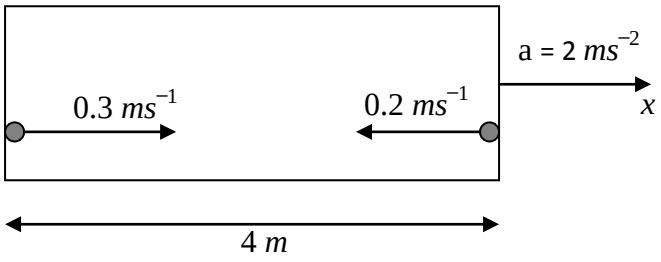
Answer: 5

17. Consider an elliptically shaped rail PQ in the vertical plane with OP = 3 *m* and OQ = 4 *m*. A block of mass 1 *kg* is pulled along the rail from P to Q with a force of 18 *N*, which is always parallel to line PQ (see the figure given). Assuming no frictional losses, the kinetic energy of the block when it reaches Q is $(n \times 10)$Joules. The value of *n* is (take acceleration due to gravity = 10 ms^{-2})



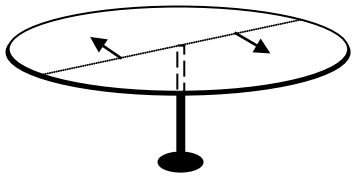
Answer : 5

18. A rocket is moving in a gravity free space with a constant acceleration of 2 ms^{-2} along +*x* direction (see figure). The length of a chamber inside the rocket is 4 *m*. A ball is thrown from the left end of the chamber in +*x* direction with a speed of 0.3 ms^{-1} relative to the rocket. At the same time, another ball is thrown in –*x* direction with a speed of 0.2 ms^{-1} from its right end relative to the rocket. The time in seconds when the two balls hit each other is



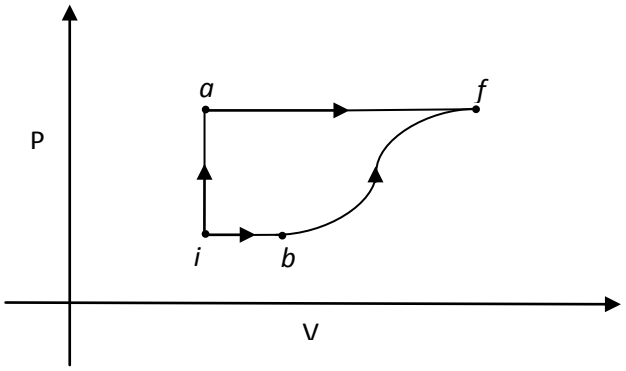
Answer : 2 OR 8

19. A horizontal circular platform of radius 0.5 *m* and mass 0.45 *kg* is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 *kg* are attached to the platform at a distance 0.25 *m* from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of 9 ms^{-1} with respect to the ground. The rotational speed of the platform in$rads^{-1}$ after the balls leave the platform is



Answer: 4

20. A thermodynamic system is taken from an initial state *i* with internal energy $U_i = 100\,J$ to the final state *f* along two different paths *iaf* and *ibf*, as schematically shown in the figure. The work done by the system along the paths *af*, *ib* and *bf* are $W_{af} = 200\,J, W_{ib} = 50\,J$ and $W_{bf} = 100\,J$ respectively. The heat supplied to the system along the path *iaf*, *ib* and *bf* are Q_{iaf}, Q_{ib} and Q_{bf} respectively. If the internal energy of the system in the state *b* is $U_b = 200\,J$ and $Q_{iaf} = 500\,J$, the ratio Q_{bf}/Q_{ib} is



Answer : 2